Concept & Outline

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ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

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ODEs: $f(x, y, y', y'', \ldots, y^{(n)}, \ldots) = 0$

n^th Order ODEs: $f(x, y, y', y'', \ldots, y^{(n)}) = 0$

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n^{th} Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1^{st} Order ODEs: $f(x, y, y') = 0$

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n[th] Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1[st] Order ODEs: $\boxed{y' = f(x, y)}$

Concept & Outline

ODEs: $f(x, y, y', y'', \ldots, y^{(n)}, \ldots) = 0$

n^th Order ODEs: $f(x, y, y', y'', \ldots, y^{(n)}) = 0$

1^st Order ODEs: $\boxed{y' = f(x, y)}$ $\{$ Initial Value Problems

Concept & Outline

ODEs: $f(x, y, y', y'', \ldots, y^{(n)}, \ldots) = 0$

n[th] Order ODEs: $f(x, y, y', y'', \ldots, y^{(n)}) = 0$

1[st] Order ODEs: $\boxed{y' = f(x, y)}$ { Initial Value Problems

Higher Order ODEs: $f(x, y, y', \ldots) = 0$

Concept & Outline

ODEs: $f(x, y, y', y'', \ldots, y^{(n)}, \ldots) = 0$

n^th Order ODEs: $f(x, y, y', y'', \ldots, y^{(n)}) = 0$

1^st Order ODEs: $\boxed{y' = f(x, y)}$ { Initial Value Problems

Higher Order ODEs: $f(x, y, y', \ldots) = 0$ { Initial Value Problems / Boundary Value Problems

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n^th Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1^st Order ODEs: $\boxed{y' = f(x, y)}$ { Initial Value Problems

Higher Order ODEs: $f(x, y, y', \dots) = 0$ { Initial Value Problems; Boundary Value Problems

Initial Value Problems: All conditions are given at a point.

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n^{th} Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1^{st} Order ODEs: $\boxed{y' = f(x, y)}$ $\{$ Initial Value Problems

Higher Order ODEs: $f(x, y, y', \dots) = 0$ $\left\{ \begin{array}{l} \text{Initial Value Problems} \\ \text{Boundary Value Problems} \end{array} \right.$

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n^th^ Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1^st^ Order ODEs: $\boxed{y' = f(x, y)}$ $\{$ Initial Value Problems

Higher Order ODEs: $f(x, y, y', \dots) = 0$ $\left\{ \begin{array}{l} \text{Initial Value Problems} \\ \text{Boundary Value Problems} \end{array} \right.$

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

System of 1^st^ Order ODEs:

$$\boxed{\begin{array}{l} y_1' = f_1(x, y_1, y_2, \dots, y_n) \\ y_2' = f_2(x, y_1, y_2, \dots, y_n) \\ \quad\vdots \\ y_n' = f_n(x, y_1, y_2, \dots, y_n) \end{array}}$$

Concept & Outline

7 ODEs: Initial Value Problems

ODEs: $f(x, y, y', y'', \ldots, y^{(n)}, \ldots) = 0$

nth Order ODEs: $f(x, y, y', y'', \ldots, y^{(n)}) = 0$

1st Order ODEs: $\boxed{y' = f(x, y)}$ $\{$ Initial Value Problems

Higher Order ODEs: $f(x, y, y', \ldots) = 0$ $\{$ Initial Value Problems; Boundary Value Problems

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

System of 1st Order ODEs:

$$\boxed{\begin{aligned} y_1' &= f_1(x, y_1, y_2, \ldots, y_n) \\ y_2' &= f_2(x, y_1, y_2, \ldots, y_n) \\ &\vdots \\ y_n' &= f_n(x, y_1, y_2, \ldots, y_n) \end{aligned}}$$

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n^th Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1^st Order ODEs: $\boxed{y' = f(x, y)}$ $\{$ Initial Value Problems

Higher Order ODEs: $f(x, y, y', \dots) = 0$ $\begin{cases} \text{Initial Value Problems} \\ \text{Boundary Value Problems} \end{cases}$

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

System of 1^st Order ODEs:

$$\boxed{\begin{aligned} y_1' &= f_1(x, y_1, y_2, \dots, y_n) \\ y_2' &= f_2(x, y_1, y_2, \dots, y_n) \\ &\vdots \\ y_n' &= f_n(x, y_1, y_2, \dots, y_n) \end{aligned}}$$

7 ODEs: Initial Value Problems

- Taylor's Method
  Calculation of Taylor series.

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n th Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1 st Order ODEs: $\boxed{y' = f(x, y)}$ { Initial Value Problems

Higher Order ODEs: $f(x, y, y', \dots) = 0$ { Initial Value Problems, Boundary Value Problems

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

System of 1 st Order ODEs:

$$\boxed{\begin{aligned} y_1' &= f_1(x, y_1, y_2, \dots, y_n) \\ y_2' &= f_2(x, y_1, y_2, \dots, y_n) \\ &\vdots \\ y_n' &= f_n(x, y_1, y_2, \dots, y_n) \end{aligned}}$$

7 ODEs: Initial Value Problems

- Taylor's Method
  Calculation of Taylor series.
- One-Step Method
  Calculation over an interval.

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n^{th} Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1^{st} Order ODEs: $\boxed{y' = f(x, y)}$ { Initial Value Problems

Higher Order ODEs: $f(x, y, y', \dots) = 0$ { Initial Value Problems, Boundary Value Problems

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

System of 1^{st} Order ODEs:

$$\boxed{\begin{aligned} y_1' &= f_1(x, y_1, y_2, \dots, y_n) \\ y_2' &= f_2(x, y_1, y_2, \dots, y_n) \\ &\vdots \\ y_n' &= f_n(x, y_1, y_2, \dots, y_n) \end{aligned}}$$

7 ODEs: Initial Value Problems

- Taylor's Method
  Calculation of Taylor series.
- One-Step Method
  Calculation over an interval.
  - Explicit Euler Method
  - Implicit Euler Method
  - Variants of Euler Method
  - Explicit Runge-Kutta Methods
  - Implicit Runge-Kutta Methods

Concept & Outline

ODEs: $f(x, y, y', y'', \ldots, y^{(n)}, \ldots) = 0$

n^{th} Order ODEs: $f(x, y, y', y'', \ldots, y^{(n)}) = 0$

1^{st} Order ODEs: $\boxed{y' = f(x, y)}$ { Initial Value Problems

Higher Order ODEs: $f(x, y, y', \ldots) = 0$ { Initial Value Problems; Boundary Value Problems

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

System of 1^{st} Order ODEs:

$$\boxed{\begin{aligned} y_1' &= f_1(x, y_1, y_2, \ldots, y_n) \\ y_2' &= f_2(x, y_1, y_2, \ldots, y_n) \\ &\vdots \\ y_n' &= f_n(x, y_1, y_2, \ldots, y_n) \end{aligned}}$$

7 ODEs: Initial Value Problems

- Taylor's Method
  Calculation of Taylor series.
- One-Step Method
  Calculation over an interval.
  - Explicit Euler Method
  - Implicit Euler Method
  - Variants of Euler Method
  - Explicit Runge-Kutta Methods
  - Implicit Runge-Kutta Methods
- Multistep Method
  Calculation over intervals.

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n

n^{th} Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1^{st} Order ODEs: $\boxed{y' = f(x, y)}$ $\{$ Initial Value Problems

Higher Order ODEs: $f(x, y, y', \dots) = 0$ $\{$ Initial Value Problems, Boundary Value Problems

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

System of 1^{st} Order ODEs:

$$\boxed{\begin{aligned} y_1' &= f_1(x, y_1, y_2, \dots, y_n) \\ y_2' &= f_2(x, y_1, y_2, \dots, y_n) \\ &\vdots \\ y_n' &= f_n(x, y_1, y_2, \dots, y_n) \end{aligned}}$$

7 ODEs: Initial Value Problems

- **Taylor's Method**
  Calculation of Taylor series.
- **One-Step Method**
  Calculation over an interval.
  - Explicit Euler Method
  - Implicit Euler Method
  - Variants of Euler Method
  - Explicit Runge-Kutta Methods
  - Implicit Runge-Kutta Methods
- **Multistep Method**
  Calculation over intervals.
  - Adams-Bashforth Methods
  - Adams-Moulton Methods
  - Predictor-Corrector Methods

Concept & Outline

ODEs: $f(x, y, y', y'', \dots, y^{(n)}, \dots) = 0$

n^{th} Order ODEs: $f(x, y, y', y'', \dots, y^{(n)}) = 0$

1^{st} Order ODEs: $\boxed{y' = f(x, y)}$ { Initial Value Problems

Higher Order ODEs: $f(x, y, y', \dots) = 0$ { Initial Value Problems / Boundary Value Problems

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

System of 1^{st} Order ODEs:

$$\boxed{\begin{aligned} y_1' &= f_1(x, y_1, y_2, \dots, y_n) \\ y_2' &= f_2(x, y_1, y_2, \dots, y_n) \\ &\vdots \\ y_n' &= f_n(x, y_1, y_2, \dots, y_n) \end{aligned}}$$

7 ODEs: Initial Value Problems

- **Taylor's Method**
  Calculation of Taylor series.
- **One-Step Method**
  Calculation over an interval.
  - Explicit Euler Method
  - Implicit Euler Method
  - Variants of Euler Method
  - Explicit Runge-Kutta Methods
  - Implicit Runge-Kutta Methods
- **Multistep Method**
  Calculation over intervals.
  - Adams-Bashforth Methods
  - Adams-Moulton Methods
  - Predictor-Corrector Methods
- **System of First Order ODEs**
  Transformation of higher order ODEs.

Concept & Outline

ODEs: $f(x, y, y', y'', \ldots, y^{(n)}, \ldots) = 0$

n^{th} Order ODEs: $f(x, y, y', y'', \ldots, y^{(n)}) = 0$

1^{st} Order ODEs: $\boxed{y' = f(x, y)}$ { Initial Value Problems

Higher Order ODEs: $f(x, y, y', \ldots) = 0$ { Initial Value Problems, Boundary Value Problems

Initial Value Problems: All conditions are given at a point.

Boundary Value Problems: All conditions are given at more than one point.

System of 1^{st} Order ODEs:

$$\begin{aligned} y_1' &= f_1(x, y_1, y_2, \ldots, y_n) \\ y_2' &= f_2(x, y_1, y_2, \ldots, y_n) \\ &\vdots \\ y_n' &= f_n(x, y_1, y_2, \ldots, y_n) \end{aligned}$$

7 ODEs: Initial Value Problems

- Taylor's Method
  Calculation of Taylor series.
- One-Step Method
  Calculation over an interval.
  - Explicit Euler Method
  - Implicit Euler Method
  - Variants of Euler Method
  - ✓ Explicit Runge-Kutta Methods
  - Implicit Runge-Kutta Methods
- Multistep Method
  Calculation over intervals.
  - Adams-Bashforth Methods
  - Adams-Moulton Methods
  - Predictor-Corrector Methods
- ✓ System of First Order ODEs
  Transformation of higher order ODEs.